Main results

Positive Summation Kernels

A family of functions $(K_n)_{n\geq 1}$ on $\mathbb{R}$ (or on $\mathbb{T}$, the circle) is called a *positive summation kernel* if the following hold:

(i) $K_n(x) \geq 0$ for all $x \in \mathbb{R}$ and each n;

(ii) $\displaystyle\int_{-\infty}^{\infty} K_n(x)\,dx = 1$ for all n;

(iii) for every $\delta > 0$,

$$\int_{|x|\geq\delta} K_n(x)\,dx \longrightarrow 0 \quad \text{as } n \to \infty.$$

Equivalently, (K_n) is an *approximate identity*: the mass of K_n concentrates near $x = 0$ as $n \to \infty$. For such kernels, the convolution $(f * K_n)(x) = \int f(x-t)K_n(t)\,dt$ satisfies

$$(f * K_n)(x) \to f(x)$$

for all x where f is continuous.

Riemann-Lebesgue Lemma

Theorem 2.2 (Riemann–Lebesgue lemma) Let f be absolutely integrable in the Riemann sense on a finite or infinite interval I. Then

$$\lim_{\lambda\to\infty} \int_I f(u)\sin(\lambda u)\,du = 0.$$

(A corresponding result holds for $\cos(\lambda u)$ or $e^{i\lambda u}$.)

Corollary for Fourier Coefficients

Corollary. Let $f \in L^1(\mathbb{T})$ (i.e., $\int_{-\pi}^{\pi} |f(x)|dx < \infty$). Its complex Fourier coefficients are:

$$\hat{f}(n) = c_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-inx}\,dx, \qquad n \in \mathbb{Z}.$$

The Fourier coefficients of any integrable function vanish at infinity:

$$\hat{f}(n) \longrightarrow 0 \quad \text{as } |n| \to \infty.$$

Proof Sketch of Corollary: The Fourier coefficient $\hat{f}(n)$ can be expressed as:

$$\hat{f}(n) = \frac{1}{2\pi}\left(\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx + i\int_{-\pi}^{\pi} f(x)\sin(-nx)\,dx\right)$$

Since $\cos(nx)$ and $\sin(nx)$ are bounded functions, the absolute integrability of f is preserved when multiplied by these functions. As $|n| \to \infty$, both the real and imaginary parts of the integral tend to zero by applying the Riemann–Lebesgue lemma (and its cosine variant) with $\lambda = n$. Therefore, $\hat{f}(n) \to 0$ as $|n| \to \infty$.

Parseval's Identity

For square-integrable functions $f, g \in L^2(\mathbb{T})$, the Parseval formulas for the squared L^2-norm and the inner product are:

$$\frac{1}{2\pi}\int_{\mathbb{T}} |f(t)|^2 dt = \sum_{n\in\mathbb{Z}} |c_n|^2, \quad \frac{1}{2\pi}\int_{\mathbb{T}} f(t)\overline{g(t)}dt = \sum_{n\in\mathbb{Z}} c_n\overline{d_n},$$

where $c_n = \hat{f}(n)$ and $d_n = \hat{g}(n)$. The latter is often called the *polarized Parseval formula.*

Criterion for $L^2(\mathbb{T})$: A function f exists in $L^2(\mathbb{T})$ if and only if its Fourier coefficients are square-summable, $\sum_{n=-\infty}^{\infty} |\hat{f}(n)|^2 < \infty$.